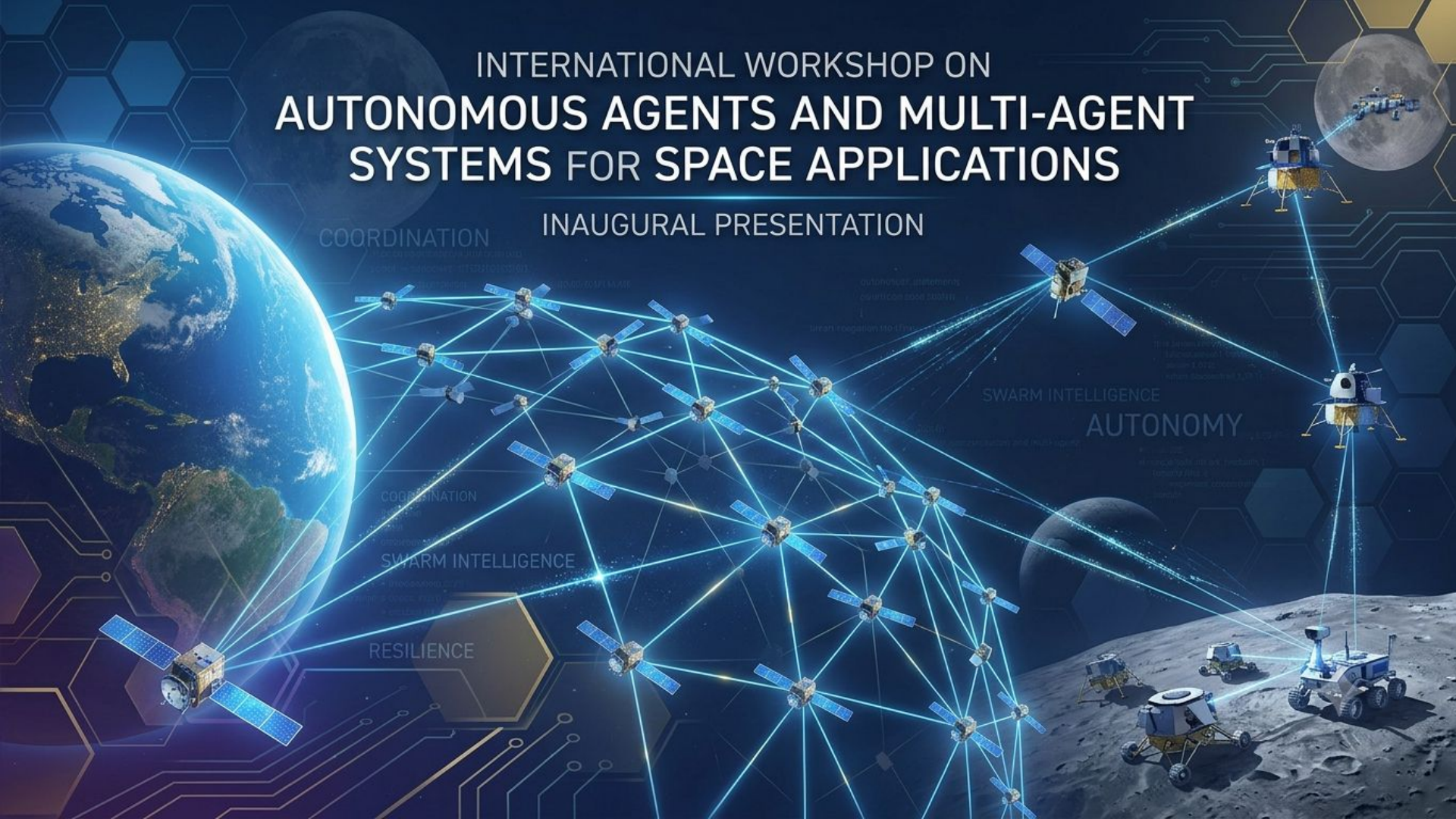


INTERNATIONAL WORKSHOP ON AUTONOMOUS AGENTS AND MULTI-AGENT SYSTEMS FOR SPACE APPLICATIONS

INAUGURAL PRESENTATION



MASSpace 2026

- Second edition (first in 2024 in Auckland)
 - <https://mas-space.github.io/aamas2024ws/>
- Multidisciplinary meeting place
- Contributions of autonomous agents and multi-agent systems to the space domain
 - more and more complex and composite systems (e.g. larger constellations, multiple-mission federations, multi-user systems, heterogeneous robotic systems)
 - more and more accurate environmental monitoring, complex requests, or richer exploration scenarios



Organizers

Steve A. Chien
NASA JPL, USA



Gauthier Picard
ONERA, France



Itai Zilberstein
CMU, USA



Programme (morning sessions)

08:45-10:15	Session 1
08:45-09:35	Invited Talk: Developing Distributed Autonomous Space Systems
09:35-09:55	<i>Large-Scale Continual Scheduling and Execution for Dynamic Distributed Satellite Constellation Observation Allocation</i>
09:55-10:15	<i>The Role of Central Nodes in Multi-Task Allocation for Distributed Space-Based Observation Systems</i>
10:15-11:00	Morning Coffee Break
11:00-12:20	Session 2
11:00-11:20	<i>Dynamic Tool Generation for Autonomous Multi-Agent Systems in Space Missions</i>
11:20-11:40	<i>Autonomous Federation of Earth Observation Constellations: An End-to-End Demonstrator based on the DOMINO Architecture</i>
11:40-12:00	<i>Solving Organizational Multi-Robot Task Allocation Problems with Consensus-based Auctions</i>
12:00-12:20	<i>Multi-Satellite Observation Tasks Dispatching and Scheduling</i>
12:30-14:00	Lunch Break

Invited Talk: Caleb A. Adams (NASA)

Developing Distributed Autonomous Space Systems

The development of distributed autonomous space systems is crucial for advancing NASA's Small Spacecraft and Distributed Systems (SSDS) program. The Distributed Spacecraft Autonomy (DSA) project at NASA's Ames Research Center aims to enable autonomous multi-agent decision-making in multi-spacecraft operations, mitigating the effects of communication constraints like latency and bandwidth. This talk will discuss recent advancements in DSA technologies and applications, including distributed resource and task management, reactive operations, and ad hoc network communications. In addition to technical implementations, we will discuss software development strategies for rapid on-orbit testing. We will also highlight successes from the Starling 1.0 mission, The Starling 1.5 mission extension, and a space traffic coordination technology demonstration that showcased autonomous satellite operations without human intervention. Our work focuses on building flight heritage and experience for low-TRL software, raising confidence in new autonomy and distributed control capabilities for next-generation space missions.



Programme (afternoon session)

14:00-15:30	Session 3
14:00-14:50	Invited Talk: A Game-Theoretic Framework for Distributed Fallback Autonomy in Satellite Mega-Constellations
14:50-15:10	<i>Satellite scheduling optimization for BepiColombo STC channel target-phase</i>
15:10-15:30	<i>Toward Automated Operational Assist in Satellite Fleet Management via Organizational MARL</i>
15:30-16:15	Afternoon Coffee Break
16:15-17:35	Session 4
16:15-16:35	<i>Decentralized Dynamic Task Allocation under Limited Communication Range</i>
16:35-16:55	<i>A queue-based approach for fine-tuning the efficiency-fairness tradeoff in distributed satellite scheduling</i>
16:55-17:15	<i>Onboard Planning and Scheduling for Autonomous Satellite Operations Towards Multi-Agent Cooperation</i>
17:15-17:35	<i>Effects of Lookahead and Communication Dynamics on Multi-Agent Potential-Field Rover Navigation</i>
17:35-17:45	Closing remarks

Invited Talk: Daniel Reynolds (MIT)

Priority-Sensitive Mission Continuity for Degraded Multi-Orbit Satellite Constellations



Satellite mega-constellations are increasingly being deployed to provide resilient global service, yet the same disruptions that motivate proliferated architectures, including large-scale outages, fragmented connectivity, stale global state, and degraded coordination, also challenge the command-and-control assumptions on which those systems depend. This talk presents a hybrid game theoretic and predictive control architecture for distributed fallback autonomy in heterogeneous multi-orbit constellations operating under disruption. The central claim is operational: when centralized or ground-mediated coordination degrades, resilience depends on coupling scalable constellation-level coordination with agent-local execution feasibility. The proposed framework combines a broadcast coordination field with onboard receding-horizon task selection, enabling satellites to continue making locally executable, mission-aware decisions without centralized reassignment. We will highlight how this approach enables constellations to respond to disruption through priority-sensitive graceful degradation rather than mission collapse, preserving operational usefulness even when coordination is partial, delayed, or unavailable. More broadly, the work aims to advance resilient distributed autonomy for future space missions by demonstrating a scalable framework for mission-aware decision-making in degraded and communications-constrained environments.

Proceedings available online



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26 May 2026, Paphos, Cyprus

<https://mas-space.github.io/aamas2026ws/#programme>

- Collected papers
- Individual papers
- Presentations will be available after the workshop

Programme Committee (many thanks!)

- Caleb A. Adams, NASA
- Alexandre Albore, ONERA
- Abigail Breitfeld, Carnegie Mellon University
- Thibault Gateau, ISAE-SUPAERO
- Alessandro Golkar, Technical University of Munich
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INAUGURAL PRESENTATION

COORDINATION

COORDINATION

SWARM INTELLIGENCE

RESILIENCE

SWARM INTELLIGENCE

AUTONOMY

Have a nice workshop!

